

Roll No.....

TCH 201
EVEN END SEMESTER EXAMINATION JUNE - 2025
B. Tech (Second Semester)
ENGINEERING CHEMISTRY

Time: Three Hours

Maximum Marks: 100

Note:

- I. All Question are compulsory
- II. Answer any two sub question among a, b & c in each main question.
- III. Total marks in each main question are twenty.
- IV. Each question carries 10 marks.

Q1. The Molecular Orbital Theory (MOT) provides a quantum mechanical description of bonding in both homo and hetero diatomic molecules. Based on MOT answer the following:

- (a). Calculate the Bond order in CO, and O_2^{2-} . Draw the molecular orbital diagram for both.

(4+4+1+1)

- (b). Explain how the band theory, derived from metallic bonding concepts, distinguishes between conductors, insulators, and semiconductors in terms of their electronic structure and conductivity.

(4+2+2+2)

- (c). Explain the energy levels involved in vibration, and electronic level transitions in spectroscopy

(10)

Q2. Hardness in water can have many negative implications if fed into boiler. Answer the following based on hardness and methods of its removal.

- (a). What are the various boiler problems faced if it is fed with hard water for steam generation?

Elaborate with impact of each problem on boiler. (6+4)

- (b). What is lime soda of softening hard water? Calculate the amount of lime (90% pure) and soda (95% pure) needed for treating 20,000 L of water containing the following impurities present per liter: $CaCl_2$ (111 mg), $MgSO_4$ (120 mg), $Ca(HCO_3)_2$ (162 mg) and $Mg(HCO_3)_2$ (146 mg) respectively. (3+7)

- (c). Among the processes known for softening hard water, which of the processes is most efficient? Compare the processes known in terms of cost involved and efficiency of process. (6+4)

Q3. The polymerization process depends on the structure of monomer, type of initiator used, and many other factors. Answer the following based on the polymerization process involved:

(a). What type of monomers leads to addition and condensation polymerization? Explain "Functionality" and give two examples of monomers for each type involved in these polymerization process. (4+2+2+2)

(b). Propose the mechanism for polymerization of ethene explaining the initiation, propagation and termination steps involved. Give some applications of polyethene. (2+2+2+2+2)

(c). What are biodegradable polymers? Outline the methods for preparation and application of Polylactic acid and polyvinyl alcohol. (2+4+4)

Q4. Renewable fuels are cleaner, sustainable, and more secure than fossil fuels. They are essential for addressing climate change, ensuring future energy supply, and protection.

(a). Explain in detail the importance of renewable energy sources and describe any two in detail. (5+5)

(b). Give the construction and working of Bomb's Calorimeter used for determination of calorific value of solid fuel. Also explain the significance of acid and fuse wire correction used. (Diagram with labelling: (6+2+2)

(c). Mention the condition when GCV is equal to NCV with the help of formula.

The following data were obtained when the coal sample containing 6.5% Hydrogen and 90% Carbon was tested in bomb calorimeter: (2+8)

Weight of crucible= 3.649g

Weight of crucible+ Fuel= 4.678g

Weight of water= 2200g

Water equivalent of calorimeter=570g

Observed rise in temperature=3°C

Latent heat of steam=587cal/g

Calculate the gross and net calorific value of a fuel.

Q5. The standard electrode potential of a half-cell is the voltage developed by it when connected to the standard hydrogen electrode, measured under standard conditions. Based on this statement answer the following:

(a). Draw a neat and labelled diagram of a Daniel cell and calculate the EMF of the electrochemical cell: (6+4)

$\text{Zn(s)} | \text{Zn}^{2+} (0.1 \text{ M}) || \text{Cu}^{2+} (1.0 \text{ M}) | \text{Cu(s)}$

Given: Standard electrode potential $E^\circ_{\text{Zn}^{2+}/\text{Zn}} = -0.76\text{V}$, $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = +0.34\text{V}$

(b). What is the electrochemical theory of Corrosion? Explain the factors affecting corrosion. (5+5).

(c). What is the Electrochemical Series? Discuss its applications in predicting the feasibility of redox reactions, displacement reactions, and corrosion protection. (4+2+2+2)